15
FRAUNHOFER DIFFRACTION BY A SINGLE OPENING

When a beam of light passes through a narrow slit, it spreads out to a certain extent into the region of the geometrical shadow. This effect, already noted and illustrated at the beginning of Chap. 13, Fig. 13B, is one of the simplest examples of *diffraction*, i.e., of the failure of light to travel in straight lines. It can be satisfactorily explained only by assuming a wave character for light, and in this chapter we shall investigate quantitatively the *diffraction pattern*, or distribution of intensity of the light behind the aperture, using the principles of wave motion already discussed.

15.1 FRESNEL AND FRAUNHOFER DIFFRACTION

Diffraction phenomena are conveniently divided into two general classes, (1) those in which the source of light and the screen on which the pattern is observed are effectively at infinite distances from the aperture causing the diffraction and (2) those in which either the source or the screen, or both, are at finite distances from the aperture. The phenomena coming under class (1) are called, for historical reasons, *Fraunhofer diffraction*, and those coming under class (2) *Fresnel diffraction*. Fraunhofer diffraction is much simpler to treat theoretically. It is easily observed in practice by rendering the light from a source parallel with a lens and focusing it on a screen with another lens placed behind the aperture, an arrangement which effectively removes the source and screen to infinity. In the observation of Fresnel diffraction, on the other hand, no lenses are necessary, but here the wave fronts are divergent instead of plane, and the theoretical treatment is consequently more complex. Only Fraunhofer diffraction will be considered in this chapter, and Fresnel diffraction in Chap. 18.

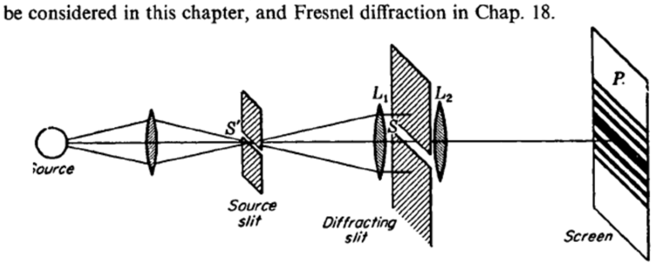

FIGURE 15A
Experimental arrangement for obtaining the diffraction pattern of a single slit; Fraunhofer diffraction.

15.2 DIFFRACTION BY A SINGLE SLIT

A slit is a rectangular aperture of length large compared to its breadth. Consider a slit S to be set up as in Fig. 15A, with its long dimension perpendicular to the plane of the page, and to be illuminated by parallel monochromatic light from the narrow slit S', at the principal focus of the lens L_1. The light focused by another lens L_2 on a screen or photographic plate P at its principal focus will form a diffraction pattern, as indicated schematically. Figure 15B(*b*) and (*c*) shows two actual photographs, taken with different exposure times, of such a pattern, using violet light of wavelength 4358 Å. The distance $S'L_1$ was 25.0 cm, and L_2P was 100 cm. The width of the slit S was 0.090 mm, and of S', 0.10 mm. If S' was widened to more than about 0.3 mm, the details of the pattern began to be lost. On the original plate, the half width d of the central maximum was 4.84 mm. It is important to notice that the width of the central maximum is *twice* as great as that of the fainter side maxima. That this effect comes under the heading of diffraction as previously defined is clear when we note that the strip drawn in Fig. 15B(*a*) is the width of the geometical image of the slit S', or practically that which would be obtained by removing the second slit and using the whole aperture of the lens. This pattern can easily be observed by ruling a single transparent line on a photographic plate and using it in front of the eye as explained in Sec. 13.2, Fig. 13E.

The explanation of the single-slit pattern lies in the interference of the Huygens secondary wavelets which can be thought of as sent out from every point on the wave front at the instant that it occupies the plane of the slit. To a first approximation, one may consider these wavelets to be uniform spherical waves, the emission of which stops abruptly at the edges of the slit. The results obtained in this way, although they give a fairly accurate account of the observed facts, are subject to certain modifications in the light of the more rigorous theory.

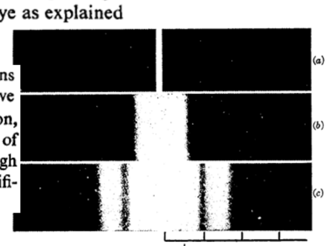

FIGURE 15B
Photographs of the single-slit diffraction pattern.

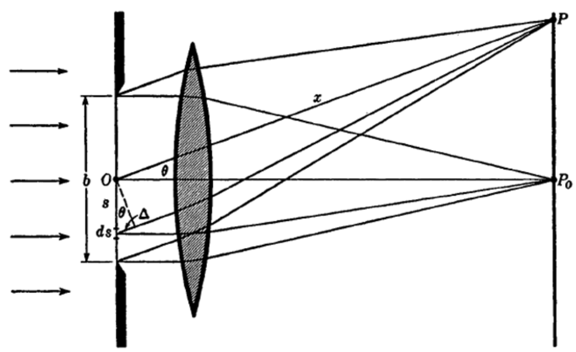

FIGURE 15C
Geometrical construction for investigating the intensity in the single-slit diffraction pattern.

Figure 15C represents a section of a slit of width b, illuminated by parallel light from the left. Let ds be an element of width of the wave front in the plane of the slit, at a distance s from the center O, which we shall call the origin. The parts of each secondary wave which travel normal to the plane of the slit will be focused at P_0, while those which travel at any angle θ will reach P. Considering first the wavelet emitted by the element ds situated at the origin, its amplitude will be directly proportional to the length ds and inversely proportional to the distance x. At P it will produce an infinitesimal displacement which, for a spherical wave, may be expressed as

$$dy_0 = \frac{a\,ds}{x} \sin(\omega t - kx)$$

As the position of ds is varied, the displacement it produces will vary in phase because of the different path length to P. When it is at a distance s below the origin, the contribution will be

$$dy_s = \frac{a\,ds}{x}\sin[\omega t - k(x + \Delta)]$$

$$= \frac{a\,ds}{x}\sin(\omega t - kx - ks\sin\theta) \qquad (15a)$$

We now wish to sum the effects of all elements from one edge of the slit to the other. This can be done by integrating Eq. (15a) from $s = -b/2$ to $b/2$. The simplest way* is to integrate the contributions from pairs of elements symmetrically placed at s and $-s$, each contribution being

$$dy = dy_{-s} + dy_s$$

$$= \frac{a\,ds}{x}[\sin(\omega t - kx - ks\sin\theta) + \sin(\omega t - kx + ks\sin\theta)]$$

By the identity $\sin\alpha + \sin\beta = 2\cos\frac{1}{2}(\alpha - \beta)\sin\frac{1}{2}(\alpha + \beta)$, we have

$$dy = \frac{a\,ds}{x}[2\cos(ks\sin\theta)\sin(\omega t - kx)]$$

which must be integrated from $s = 0$ to $b/2$. In doing so, x may be regarded as constant, insofar as it affects the amplitude. Thus

$$y = \frac{2a}{x}\sin(\omega t - kx)\int_0^{b/2}\cos(ks\sin\theta)\,ds$$

$$= \frac{2a}{x}\left[\frac{\sin(ks\sin\theta)}{k\sin\theta}\right]_0^{b/2}\sin(\omega t - kx)$$

$$= \frac{ab}{x}\frac{\sin(\frac{1}{2}kb\sin\theta)}{\frac{1}{2}kb\sin\theta}\sin(\omega t - kx) \qquad (15b)$$

The resultant vibration will therefore be a simple harmonic one, the amplitude of which varies with the position of P, since the latter is determined by θ. We may represent its amplitude as

- $$A = A_0\frac{\sin\beta}{\beta} \qquad (15c)$$

where $\beta = \frac{1}{2}kb\sin\theta = (\pi b\sin\theta)/\lambda$ and $A_0 = ab/x$. The quantity β is a convenient variable, which signifies one-half the phase difference between the contributions coming from opposite edges of the slit. The intensity on the screen is then

- $$I \approx A^2 = A_0{}^2\frac{\sin^2\beta}{\beta^2} \qquad (15d)$$

If the light, instead of being incident on the slit perpendicular to its plane, makes an angle i, a little consideration will show that it is merely necessary to replace the above expression for β by the more general expression

- $$\beta = \frac{\pi b(\sin i + \sin\theta)}{\lambda} \qquad (15e)$$

15.3 FURTHER INVESTIGATION OF THE SINGLE-SLIT DIFFRACTION PATTERN

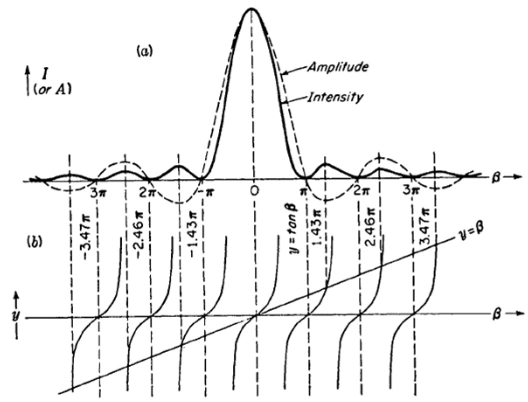

FIGURE 15D
Amplitude and intensity contours for Fraunhofer diffraction of a single slit, showing positions of maxima and minima.

In Fig. 15D(*a*) graphs are shown of Eq. (15c) for the *amplitude* (dotted curve) and Eq. (15*d*) for the *intensity*, taking the constant A_0 in each case as unity. The intensity curve will be seen to have the form required by the experimental result in Fig. 15B.

The maximum intensity of the strong central band comes at the point P_0 of Fig. 15C, where evidently all the secondary wavelets will arrive in phase because the path difference $\Delta = 0$. For this point $\beta = 0$, and although the quotient $(\sin \beta)/\beta$ becomes indeterminate for $\beta = 0$, it will be remembered that $\sin \beta$ approaches β for small angles and is equal to it when β vanishes. Hence for $\beta = 0$, $(\sin \beta)/\beta = 1$. We now see the significance of the constant A_0. Since for $\beta = 0$, $A = A_0$, it represents the amplitude when all the wavelets arrive in phase. $A_0{}^2$ is then the value of the maximum intensity, at the center of the pattern. From this *principal maximum* the intensity falls to zero at $\beta = \pm\pi$, then passes through several *secondary maxima*, with equally spaced points of zero intensity at $\beta = \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$, or in general $\beta = m\pi$. The secondary maxima do not fall halfway between these points, but are displaced toward the center of the pattern by an amount which decreases with increasing m. The exact values of β for these maxima can be found by differentiating Eq. (15c) with respect to β and equating to zero. This yields the condition

$$\tan \beta = \beta$$

The values of β satisfying this relation are easily found graphically as the intersections of the curve $y = \tan \beta$ and the straight line $y = \beta$. In Fig. 15D(b) these points of intersection lie directly below the corresponding secondary maxima.

The intensities of the secondary maxima can be calculated to a very close approximation by finding the values of $(\sin^2 \beta)/\beta^2$ at the halfway positions, i.e., where $\beta = 3\pi/2, 5\pi/2, 7\pi/2, \ldots$. This gives $4/9\pi^2$, $4/25\pi^2$, $4/49\pi^2, \ldots$, or $1/22.2$, $1/61.7$, $1/121, \ldots$, of the intensity of the principal maximum.

A very clear idea of the origin of the single-slit pattern is obtained by the following simple treatment. Consider the light from the slit of Fig. 15E coming to the point P_1 on the screen, this point being just one wavelength farther from the upper edge of the slit than from the lower. The secondary wavelet from the point in the slit adjacent to the upper edge will travel approximately $\lambda/2$ farther than that from the point at the center, and so these two will produce vibrations with a phase difference of π and will give a resultant displacement of zero at P_1. Similarly the wavelet from the next point below the upper edge will cancel that from the next point below the center, and we can continue this pairing off to include all points in the wave front, so that the resultant effect at P_1 is zero. At P_3 the path difference is 2λ, and if we divide the slit into four parts, the pairing of points again gives zero resultant, since the parts cancel in pairs. For the point P_2, on the other hand, the path difference is $3\lambda/2$, and we divide the slit into thirds, two of which will cancel, leaving one third to account for the intensity at this point. The resultant amplitude at P_2 is, of course, not even approximately one-third that at P_0, because the phases of the wavelets from the remaining third are not by any means equal.

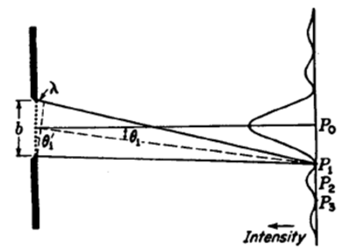

FIGURE 15E
Angle of the first minimum of the single-slit diffraction pattern.

The above method, though instructive, is not exact if the screen is at a finite distance from the slit. As Fig. 15E is drawn, the shorter broken line is drawn to cut off equal distances on the rays to P_1. It will be seen from this that the path difference to P_1 between the light coming from the upper edge and that from the center is slightly greater than $\lambda/2$ and that between the center and lower edge slightly less than $\lambda/2$. Hence the resultant intensity will not be zero at P_1 and P_3, but it will be more nearly so the greater the distance between slit and screen or the narrower the slit. This corresponds to the transition from Fresnel diffraction to Fraunhofer diffraction. Obviously, with the relative dimensions shown in the figure, the geometrical shadow of the slit would considerably widen the central maximum as drawn. Just as was true with Young's experiment (Sec. 13.3), when the screen is at infinity, the relations become simpler. Then the two angles θ_1 and θ'_1 in Fig. 15E become exactly equal, i.e., the two broken lines are perpendicular to each other, and $\lambda = b \sin \theta_1$ for the first minimum corresponding to $\beta = \pi$. This gives

- $$\sin \theta_1 = \frac{\lambda}{b} \qquad \text{(15f)}$$

In practice θ_1 is usually a very small angle, so we may put the sine equal to the angle. Then

- $$\theta_1 = \frac{\lambda}{b} \qquad \text{(15g)}$$

a relation which shows at once how the dimensions of the pattern vary with λ and b. The *linear* width of the pattern on a screen will be proportional to the slit-screen distance, which is the focal length f of a lens placed close to the slit. The linear distance d between successive minima corresponding to the angular separation $\theta_1 = \lambda/b$ is thus

$$d = \frac{f\lambda}{b}$$

15.4 GRAPHICAL TREATMENT OF AMPLITUDES. THE VIBRATION CURVE

Now, using the fact that the resultant amplitude and phase may be found by the vector addition of the individual amplitudes making angles with each other equal to the phase difference, a vector diagram like that shown in Fig. 15F(b) may be drawn. Each of the nine equal amplitudes a is inclined a tan angle δ with the preceding one, and their vector sum A is the resultant amplitude required. Now suppose that instead of dividing the slit into nine elements, we had divided it into many thousand or, in the limit, an infinite number of equal elements. The vectors a would become shorter, but at the same time δ would decrease in the same proportion, so that in the limit our vector diagram would approach the arc of a circle, shown as in (b'). The resultant amplitude A is still the same and equal to the length of the chord of this arc. Such a continuous curve, representing the addition of infinitesimal amplitudes, we shall refer to as a *vibration curve*.

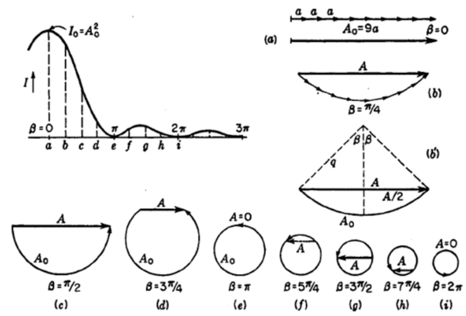

FIGURE 15F
Graphical treatment of amplitudes in single-slit diffraction.

15.5 RECTANGULAR APERTURE

In the preceding sections the intensity function for a slit was derived by summing the effects of the spherical wavelets originating from a linear section of the wave front by a plane perpendicular to the length of the slit, i.e., by the plane of the page in Fig. 15C. Nothing was said about the contributions from parts of the wave front out of this plane. A more thorough mathematical investigation, involving a double integration over both dimensions of the wave front,* shows, however, that the above result is correct when the slit is very long compared to its width. The complete treatment gives, for a slit of width b and length l, the following expression for the intensity:

$$I \approx b^2 l^2 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 \gamma}{\gamma^2} \qquad (15h)$$

where $\beta = (\pi b \sin \theta)/\lambda$, as before, and $\gamma = (\pi l \sin \Omega)/\lambda$. The angles θ and Ω are measured from the normal to the aperture at its center, in planes through the normal parallel to the sides b and l, respectively. The diffraction pattern given by Eq. (15h) when b and l are comparable with each other is illustrated in Fig. 15G.

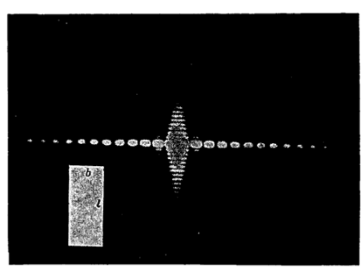

FIGURE 15G
Diffraction pattern from a rectangular opening.

$$I \approx b^2 l^2 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 \gamma}{\gamma^2} \qquad (15h)$$

where $\beta = (\pi b \sin \theta)/\lambda$, as before, and $\gamma = (\pi l \sin \Omega)/\lambda$.
the sides b and l,

Now for a slit having l very large, the factor $(\sin^2 \gamma)/\gamma^2$ in Eq. (15h) is zero for all values of Ω except extremely small ones. This means that the diffraction pattern will be limited to a line on the screen perpendicular to the slit and will resemble a section of the central horizontal line of bright spots in Fig. 15G. We do not ordinarily observe such a line pattern in diffraction by a slit, because its observation requires the use of a *point source*. In Fig. 15A the primary source was a slit S', with its long dimension perpendicular to the page. In this case, each point of the source slit forms a line pattern, but these fall adjacent to each other on the screen, adding up to give a pattern like Fig. 15B. If we were to use a slit source with the rectangular aperture of Fig. 15G, the slit being parallel to the side l, the result would be the summation of a number of such patterns, one above the other, and would be identical with Fig. 15B.

15.6 RESOLVING POWER WITH A RECTANGULAR APERTURE

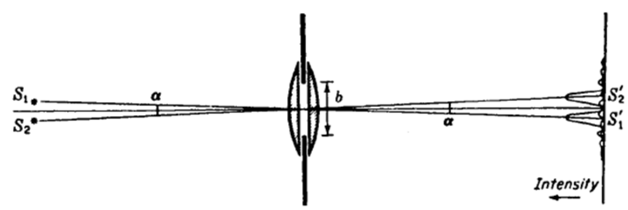

FIGURE 15H
Diffraction images of two slit sources formed by a rectangular aperture.

By the resolving power of an optical instrument we mean its ability to produce separate images of objects very close together. Using the laws of geometrical optics, one designs a telescope or a microscope to give an image of a point source which is as small as possible. However, in the final analysis, it is the diffraction pattern that sets a theoretical upper limit to the resolving power. We have seen that whenever parallel light passes through any aperture, it cannot be focused to a point image but instead gives a diffraction pattern in which the central maximum has a certain finite width, inversely proportional to the width of the aperture. The images of two objects will evidently not be resolved if their separation is much less than the width of the central diffraction maximum.

Figure 15H shows two plano-convex lenses (equivalent to a single double-convex lens) limited by a rectangular aperture of vertical dimension b. Two narrow slit sources S_1 and S_2 perpendicular to the plane of the figure form real images S_1' and S_2' on a screen. Each image consists of a single-slit diffraction pattern for which the intensity distribution is plotted in a vertical direction. The angular separation α of the central maxima is equal to the angular separation of the sources, and with the value shown in the figure is adequate to give separate images. The condition illustrated is that in which each principal maximum falls exactly on the second minimum of the adjacent pattern. This is the smallest possible value of α which will give zero intensity between the two strong maxima in the resultant pattern. The angular separation from the center to the second minimum in either pattern then corresponds to $\beta = 2\pi$ (see Fig. 15D), or $\sin\theta \approx \theta = 2\lambda/b = 2\theta_1$. As α is made smaller than this, and the two images move closer together, the intensity between the maxima will rise, until finally no minimum remains at the center. Figure 15I illustrates this by showing the resultant curve (heavy line) for four different values of α. In each case the resultant pattern has been obtained by merely adding the intensities due to the separate patterns (dotted and light curves), as was done in the case of the Fabry-Perot fringes (Sec. 14.12).

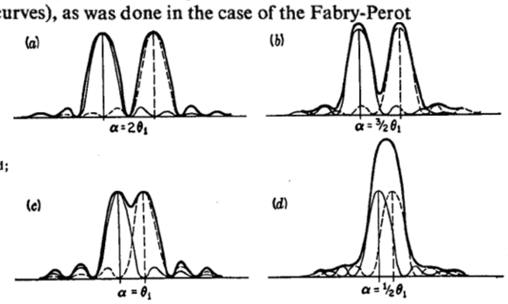

FIGURE 15I
Diffraction images of two slit sources: (a) and (b) well resolved; (c) just resolved; (d) not resolved.

Inspection of this figure shows that it would be impossible to resolve the two images if the maxima were much closer than $\alpha = \theta_1$, corresponding to $\beta = \pi$. At this separation the maximum of one pattern falls exactly on the first minimum of the other, so that the intensities of the maxima in the resultant pattern are equal to those of the separate maxima.

To find the intensity at the center of the resultant minimum for diffraction fringes separated by θ_1, we note that the curves cross at $\beta = \pi/2$ for either pattern and

$$\frac{\sin^2\beta}{\beta^2} = \frac{4}{\pi^2} = 0.4053$$

the intensity of either relative to the maximum. The sum of the contributions at this point is therefore 0.8106, which shows that the intensity of the resultant pattern drops almost to four-fifths of its maximum value. This change of intensity is easily visible to the eye, and in fact a considerably smaller change could be seen, or at least detected with a sensitive intensity-measuring instrument such as a microphotometer. However, the depth of the minimum changes very rapidly with separation in this region, and in view of the simplicity of the relations in this particular case, it was decided by Rayleigh to arbitrarily fix the separation $\alpha = \theta_1 = \lambda/b$ as the criterion for resolution of two diffraction patterns. This quite arbitrary choice is known as *Rayleigh's criterion.* The angle θ_1 is sometimes called the *resolving power* of the aperture b, although the ability to resolve increases as θ_1 becomes smaller. A more appropriate designation for θ_1 is the *minimum angle of resolution.*

15.8 CIRCULAR APERTURE

The diffraction pattern formed by plane waves from a point source passing through a circular aperture is of considerable importance as applied to the resolving power of telescopes and other optical instruments. Unfortunately it is also a problem of considerable difficulty, since it requires a double integration over the surface of the aperture similar to that mentioned in Sec. 15.5 for a rectangular aperture. The problem was first solved by Airy* in 1835, and the solution is obtained in terms of Bessel functions of order unity. These must be calculated from series expansions, and the most convenient way to express the results for our purpose will be to quote the actual figures obtained in this way (Table 15B).

The diffraction pattern as illustrated in Fig. 15K(*a*) consists of a bright central disk, known as *Airy's disk*, surrounded by a number of fainter rings. Neither the disk nor the rings are sharply limited but shade gradually off at the edges, being separated by circles of zero intensity. The intensity distribution is very much the same as that which would be obtained with the single-slit pattern illustrated in Fig. 15E by rotating it about an axis in the direction of the light and passing through the principal maximum. The dimensions of the pattern are, however, appreciably different from those in a single-slit pattern for a slit of width equal to the diameter of the circular aperture. For the single-slit pattern, the angular separation θ of the minima from the center was found in Sec. 15.3 to be given by $\sin\theta \approx \theta = m\lambda/b$, where m is any whole number, starting with unity. The dark circles separating the bright ones in the pattern from a circular aperture can be expressed by a similar formula if θ is now the angular semidiameter of the circle, but in this case the numbers m are not integers. Their numerical values as calculated by Lommel† are given in Table 15B, which also includes the values of m for the maxima of the bright rings and data on their intensities.

* Sir George Airy (1801–1892). Astronomer Royal of England from 1835 to 1881. Also known for his work on the aberration of light (Sec. 19.11). For details of the solution here referred to, see T. Preston, "Theory of Light," 5th ed., pp. 324–327, Macmillan & Co., Ltd., London, 1928.

† E. V. Lommel, *Abh. Bayer. Akad. Wiss.*, **15**:531 (1886).

Table 15B

	Circular aperture			Single slit	
Ring	m	I_{max}	I_{total}	m	I_{max}
Central maximum	0	1	1	0	1
First dark	1.220			1.000	
Second bright	1.635	0.01750	0.084	1.430	0.0472
Second dark	2.233			2.000	
Third bright	2.679	0.00416	0.033	2.459	0.0165
Third dark	3.238			3.000	
Fourth bright	3.699	0.00160	0.018	3.471	0.0083
Fourth dark	4.241			4.000	
Fifth bright	4.710	0.00078	0.011	4.477	0.0050
Fifth dark	5.243			5.000	

The column headed I_{max} gives the relative intensities of the maxima, while that headed I_{total} is the total amount of light in the ring, relative to that of the central disk. For comparison, the values of m and I_{max} for the straight bands of the single-slit pattern are also included.

(*a*)

(*b*)

(*c*)

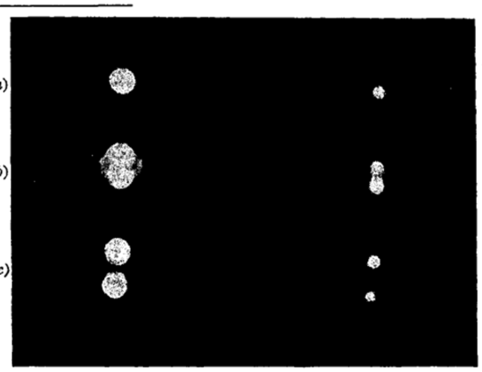

FIGURE 15K
Photographs of diffraction images of point sources taken with a circular aperture: (*a*) one source; (*b*) two sources just resolved; (*c*) two sources completely resolved.